

# POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year: 2020  
 Programme: BCA Full Marks: 100  
 Course: Mobile Application Development Pass Marks: 45  
 Technology Time: 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.  
 The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define API level. Explain in Detail how android code execution works? Define ADB command line interface. 8  
 b) Define inheritance and its types. Why multiple inheritance is not supported in java? 7
2. a) How do you make an activity default? Write down a simple program to pass data bundle from Login Activity to Profile Activity? 8  
 b) Define layout, view and view group in android. How can we declare layout in an android? How can we define styles in android? 7
3. a) Explain following: 8  
 i. `android:layout_weight="1"`  
 ii. `android:layout_gravity="right"`  
 iii. `android:text="@string/to"`  
 iv. `android:orientation="vertical"`  
 v. `CardView`  
 vi. `Spinner`  
 b) How do you make an activity default? Write down a simple program to pass data bundle from Login Activity to Profile Activity? 7
4. a) What is Alarm Manager? What are the types of alarm? 7  
 b) Define toast, toast notification, snack bar message. How does toast notification differ from status bar notification? Write a sample code to display a toast notification when the user registration is successful? 8
5. a) Define Multitouch and gestures. How it is handled in android? Explain with example. 7

- b) Explain Tweened and Interpolated view animation with sample code. 8
6. a) How do you publish application in play store? 8  
 b) Define SQLite Database. Write a sample java code to create a database named "students" in a public class StudentOpenHelper extending SQLiteOpenHelper. 7
7. Write short notes on any two: 2x5  
 a) Signing in debugging and release mode  
 b) Debug apk vs release apk  
 c) Android App Monetization

## POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2019

Programme: BCA

Full Marks: 100

Course: Mobile Application Development  
Technology

Pass Marks: 45

Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Attempt all the questions.*

1. a) Explain Mobile Operating System and its structure with a diagram. 8  
b) Explain Dalvik Virtual Machine in Details? Differentiate between JIT and AOT. 7
2. a) What do you understand by Intent? Write down its syntax? Write down the program to switch from activity "Login Activity" to another activity "Dashboard Activity". 8  
b) What is activity in Android? Explain the Android activity lifecycle with diagram. 7
3. a) Explain JVM. Why java is platform independent? 7  
b) What do you mean by DPI? How can you achieve multiple screen support in our Android application? Explain. 8
4. a) Write an xml for login layout with username, password and login button using Linear Layout. 7  
b) Describe processes, threads and broadcast Receiver class in android. 8
5. a) What is alarm Manager? What are the types of alarm? ~~elapsed~~ <sup>fire</sup> 7  
b) ~~Describe multitouch~~ gesture in android? List and explain some of the events generated during multitouch. 8
6. a) Define Sqlite Database. Write a sample sqlite code to create a database named "dictionary" in a public class DictionaryOpenHelper extending SQLiteOpenHelper. 8  
b) Explain the steps on publishing an android app in Google Play Store. 7
7. Write short notes on **any two**: 2×5
  - a) Shared Preferences
  - b) User Notification
  - c) Inheritance in Java

# POKHARA UNIVERSITY

Semester: Fall

Level: Bachelor

Year: 2017

Program: B.C. A.

Full Marks: 100

Course: Android Development Programming Technology Time: 3hrs

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

**Attempt all questions.**

1 a) What is android? Write brief history of Android. 8

b) List android API versions. What is the latest Android API version? What API version should your app would need to support maximum users? Explain. 7

2 a) What do you understand by Intent? Write down its Syntax. Write down the program to switch from activity "LoginActivity" to another activity "DashboardActivity". 8

b) What is activity in Android? Explain the Android activity life-cycle with diagram. 7

3 a) What do you understand by views in Android? Explain the following. 8

i. EditText

ii. Spinner

iii. Progress Bar

iv. Orientation in Linear Layout

b) How to work with multiple screens in Android? Explain with use cases as possible 7

4 a) Write down the advantages and disadvantages of Android operating System. Explain alert Dialog Box. 8

b) Explain the following properties of views 7

- i. android:visibility="gone"
- ii. android:text="@string/Login"
- iii. android:layout\_width="match\_parent"
- iv. android:hint="@string/Address"
- v. android:layout\_height="wrap\_content"
- vi. android:textColor="@color/black"
- vii. android:inoutType="phone"
- viii. android:id="@+id/etxtMobileNumber"
- ix. android:gravity="center"
- x. android:background="@color/green"

5 a) Write down the xml for following using relative layout. 8

Enter first name  
Enter last name  
Desired username  
Password  
Re-enter password  
Enter valid email is (optional)  
Register

b) Which would you prefer database or shared preferences for persistence storage.

6 a) Write code for button click listener to switch to another activity. 7

b) What is Toast Notification? How does it differ from status bar Notification? Write a simple code to display toast Notification when an activity is started. 8

7 Write short note on (**any two**) 5x2 = 10

a) Google Paly Services

- b) Debug apk vs Release apk
- c) Android App Monetization

## POKHARA UNIVERSITY

Semester: Spring

Level: Bachelor

Year: 2016

Program: B.C. A.

Full Marks: 100

Course: Android Development Programming Technology    Time: 3hrs

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

**Attempt all questions.**

- 1 a) What are different mobile Oses? Explain in brief one of the popular mobile OS with its architecture. 1+7
- b) Draw the architecture diagram of Android Runtime. 7
- 2 a) Differentiate between class and object with real world scenario, create class, instantiate object and explain them. 2+6
- b) What are the features of OOPs. How do you achieve polymorphism in Java? 2+5
- 3 a) What is intent in android? What are different kinds of intents supported by android? 3+5
- b) Design a basic android manifest file and explain the use of it. 7
- 4 a) How do you create the User Interface in native Android? Explain how these interfaces are bounded with activity? 2+5
- b) Give a brief overview of multiscreen support. What are the underlying factors that needs in consideration while designing multiscreen support?

OR

Write a program (activity, layout) in native android for “signup process” with assumption from any social media. 7

5 a) What is notification? Explain the importance of thread in notification? 3+5

OR

What is push notification? Explain the Google Cloud Messaging System.

b) What is multi touch gesture in android? List and explain some of the events generated during multitouch. 2+5

6 a) What is shared preferences? Explain inbuilt database in android. 3+5

b) List and Explain the steps involved in publishing an app in play store starting from source code. 7

7 Write short notes on: (Any two) 2x5 =10

- a) ADB
- b) Fragment
- c) AsyncTask